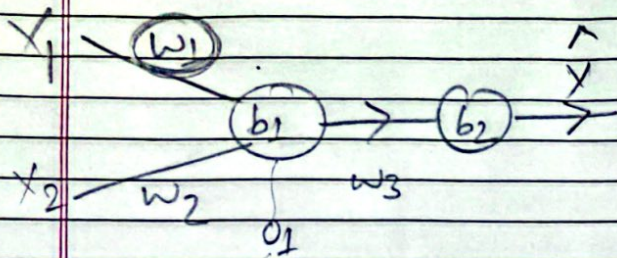


Dying ReLU Problem, Dead Neuron

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$$z_1 = w_1 x_1 + w_2 x_2 + b_1$$

$$o_1 = \max(0, z_1)$$

$$= 0 \text{ (assuming to verify)}$$

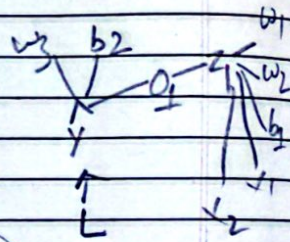
Then, $\frac{\partial o_1}{\partial z_1} = 0$

We know,

$$w_1 = w_1 - \eta \frac{\partial L}{\partial w_1}$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial o_1} \times \frac{\partial o_1}{\partial z_1} \times \frac{\partial z_1}{\partial w_1}$$

$$= 0$$



$$w_1 = w_{old} \text{ (Every time)}$$

Analysing problem,

$$o_1 = \max(0, z_1)$$

If $z_1 = w_1 x_1 + w_2 x_2 + b_1 < 0$

So, 1) High learning rate

$$w_1 = w_1 - \eta \frac{\partial L}{\partial w_1}$$

Big negative value

= -ve value

So, o_1 becomes 0 & kept

2) High negative bias

$$z_1 = w_1 x_1 + w_2 x_2 + b_1 < \text{negative}$$

then, $z_1 < 0$

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